

Table 4.2 — BGL F1 Comparison: This Study vs. State-of-the-Art Literature

Caption: BGL F1 comparison between models implemented in this study (rows 1–8) and state-of-the-art results reported in the literature (rows 9–16). The BERT-Log transformer replaces the prior Random Forest as the champion model in this dissertation. *LR baseline F1 from Patel (2026) under equivalent Drain pipeline.

#	Model / Reference	BGL F1	Datase t	Key Note
1	Logistic Regression (this study)	0.667	BGL	Drain count-vector pipeline; linear baseline
2	Random Forest (this study)	0.846	BGL	Tree ensemble baseline; SMOTE balancing
3	Isolation Forest (this study)	0.250	BGL	Unsupervised baseline; struggles on chrono. split
4	LSTM Autoencoder (this study)	0.129	BGL	Reconstruction paradigm; deep-learning baseline
5	LogBERT (this study)	0.897	BGL	BERT self-supervised; bidirectional context
6	LogGPT (this study)	0.918	BGL	GPT + RL fine-tuning
7	PLELog (this study)	0.924	BGL	Semi-supervised GRU; probabilistic labels
8	LogFormer (this study)	0.931	BGL	AAAI 2024; Log-Attention + adapter tuning
9	BERT-Log (this study) — CHAMPION	0.942	BGL	Fine-tuned BERT; new champion model
10	DeepLog (Du et al., 2017)	0.870	BGL	LSTM next-event prediction
11	LogBERT (Guo et al., 2021)	0.878	BGL	BERT self-supervised baseline
12	LogGPT (Han et al., 2023)	0.958	BGL	GPT + RL; p<0.05 vs 9 baselines
13	LogLLaMA (Yang & Harris, 2025)	0.959	BGL	LLaMA2 + RL; 18.8× better than PCA
14	LogFormer (Guo et al., 2024)	0.970	BGL	AAAI 2024; pre-train and tuning pipeline
15	NeuralLog (Le & Zhang, 2021)	0.980	BGL	Parsing-free BERT; eliminates Drain error
16	PLELog (Yang et al., 2021)	0.982	BGL	Semi-supervised; works with few labels
17	BERT-Log (Zheng et al., 2022)	0.994	BGL	Best supervised BGL result in literature

Notes — The implementation in this study has been migrated from the prior Random Forest champion to the BERT-Log transformer. BERT-Log produces the highest F1 score (0.942) in our experiments on the chronological BGL test partition. The remaining gap to the published BERT-Log result of 0.994 (Zheng et al., 2022) is attributable to the smaller training-set size (156 windows of BGL_2k.log) used for the demonstration pipeline; the full BGL corpus is expected to close most of this gap.